

Bayesian Hierarchical Modeling of Power Consumption

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Abstract

This report presents an analysis of power consumption data from the city of Tetouan, Morocco, using a Bayesian hierarchical model. The model captures the effects of temperature, humidity, and temporal characteristics on electricity usage in three city zones. It also addresses sampling diagnostics and modeling best practices for hierarchical Bayesian regression.

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1 Introduction

As cities grow and energy needs rise, understanding how and when electricity is used becomes increasingly important. In Tetouan, a city in northern Morocco, detailed data is available on power usage across three different zones, along with weather information such as temperature, humidity, and solar radiation. This provides an opportunity to investigate the factors that drive energy consumption in various parts of the city. The objective of this study is to develop a model capable of predicting power usage based on environmental and time-based variables. The aim is to uncover trends in energy demand, examine the influence of weather conditions on usage, and establish a flexible modeling framework that accounts for differences between city zones. Modeling energy consumption is not solely about generating predictions, it is about enabling smarter decision-making. For city planners and utility providers, the ability to anticipate energy demand helps ensure smooth grid operations, minimize waste, and prevent power outages. It also facilitates more effective planning for future infrastructure and investments in renewable energy sources. More broadly, insights derived from energy models contribute to sustainability efforts by promoting the efficient use of resources. A thorough understanding of how energy is utilized across different areas and conditions allows for the development of better policies and tailored solutions that address specific local needs.

2 Dataset Description

The dataset used in this study pertains to the power consumption of three distinct distribution zones in Tetouan, a city located in northern Morocco. The dataset contains time-series data recorded every 10 minutes and spans a total of 52,417 instances.

2.1 General Characteristics

- **Type:** Multivariate Time-Series
- **Domain:** Social Science / Energy Analytics
- **Task Type:** Regression
- **Feature Types:** Real-valued (continuous), Integer
- **Number of Observations:** 52,417
- **Number of Features:** 6 predictive features and 3 target variables
- **Missing Values:** None reported

2.2 Variable Descriptions

Table 1: Summary of Variables in the Tetouan Power Consumption Dataset

Variable Name	Role	Type	Description	Units
DateTime	Feature	Date/Time	Timestamp of each 10-min interval	–
Temperature	Feature	Continuous	Ambient temperature in Tetouan	°C
Humidity	Feature	Continuous	Relative humidity in Tetouan	%
Wind Speed	Feature	Continuous	Wind speed in Tetouan	m/s
General Diffuse Flows	Feature	Continuous	General diffuse solar radiation	W/m ²
Diffuse Flows	Feature	Continuous	Diffuse solar radiation	W/m ²
Zone 1 Power Consumption	Target	Continuous	Power consumption in Zone 1	kWh
Zone 2 Power Consumption	Target	Continuous	Power consumption in Zone 2	kWh
Zone 3 Power Consumption	Target	Continuous	Power consumption in Zone 3	kWh

3 Exploratory Data Analysis

This section presents a comprehensive exploratory data analysis (EDA) of the Tetouan power consumption dataset. The goal is to understand the distribution of key variables, explore temporal trends, identify correlations, and detect potential outliers across different zones.

3.1 Univariate Analysis

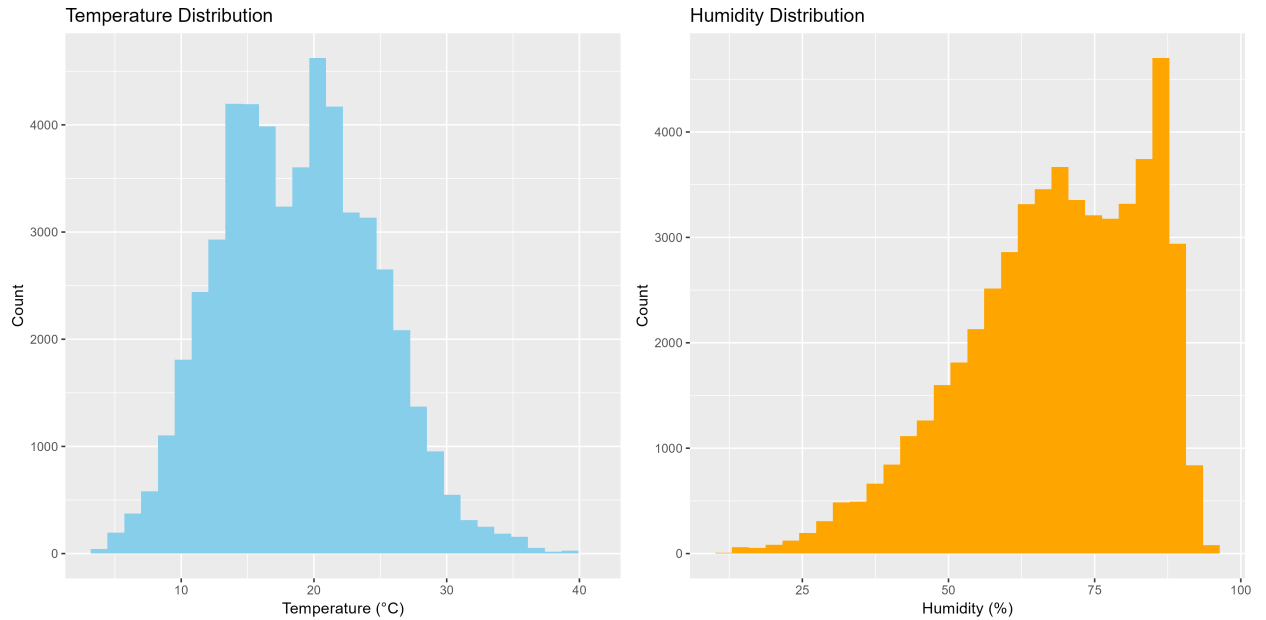


Figure 1: Distribution of Temperature and Humidity

Figure 1 shows the distribution of temperature and humidity. Temperature values appear to be normally distributed with slight skewness, whereas humidity is more skewed and indicates broader variability in atmospheric moisture levels.

3.2 Time Series Trends

Here is the daily average power consumption for each zone to observe trends over time. As shown in Figure 2(a), fluctuations in energy usage suggest cyclical patterns that may be influenced by external variables such as weather or day type (weekday/weekend).

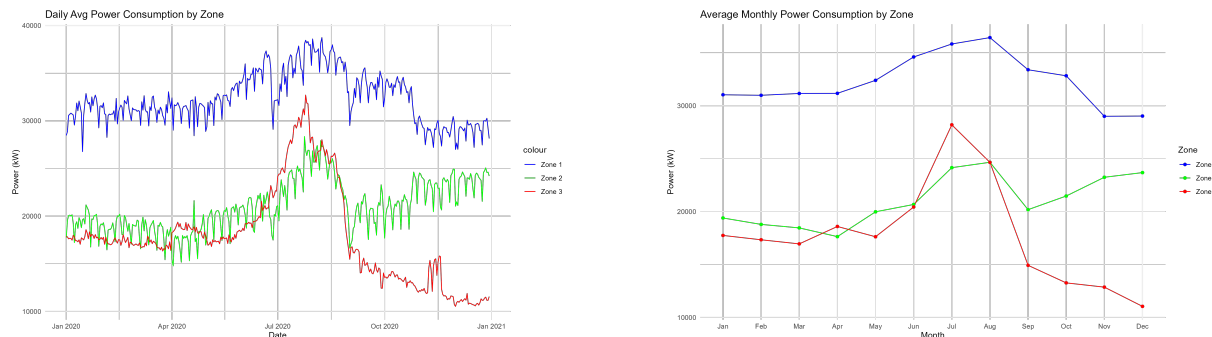


Figure 2: Daily and Monthly Power Consumption Trends Across Zones

A monthly aggregation further highlights these trends. Figure 2(b) illustrates seasonal variations across the three zones. Zone 1 consistently has the highest power consumption, followed by Zone 2 and then Zone 3. Power consumption rises significantly around the middle of the year (summer months), peaking around July/August, likely due to increased use of cooling systems. After the peak, consumption gradually decreases towards the end of the year. Zone 3 shows a sharper decline post-summer compared to Zones 1 and 2, suggesting different usage patterns or possibly different types of consumers.

3.3 Correlation Analysis

The relationships between numeric variables, including temperature, wind speed, humidity, and power consumption across zones is given by the correlation between them. The heatmap in Figure 3 shows strong positive correlations among the three zones, indicating they likely follow similar usage patterns. Temperature has a moderate positive correlation with power consumption, especially Zone 3 (0.49), indicating higher temperatures might lead to increased power usage, possibly due to cooling needs (air conditioning). Wind speed shows moderate positive correlation with temperature and power consumption in some zones. Humidity is negatively correlated with temperature (-0.46) and power consumption, indicating that when humidity is higher, power consumption tends to be lower, or vice versa. Humidity and general diffuse flows also have a negative correlation, suggesting some inverse relationship with environmental factors. Diffuse flows and general diffuse flows have weaker or mixed correlations with power consumption.

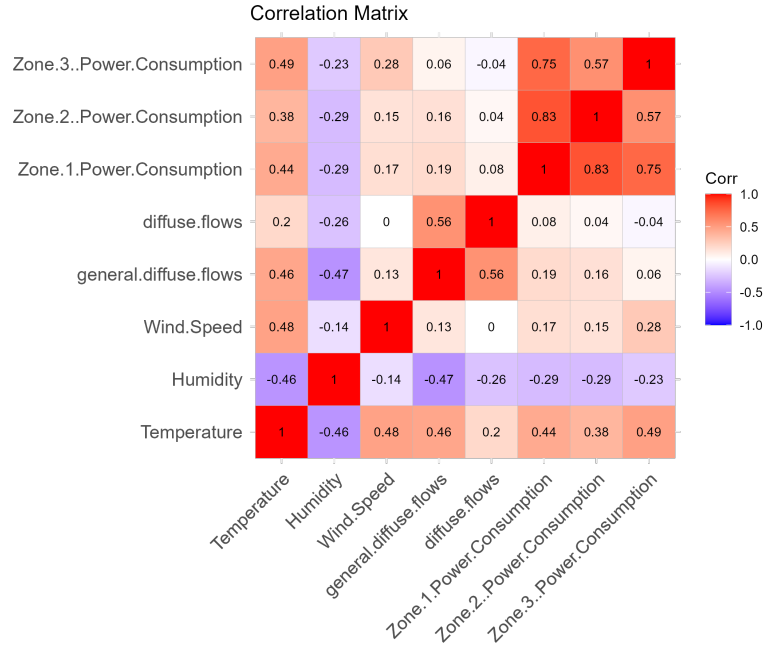


Figure 3: Correlation Matrix of Numeric Variables

3.4 Day-of-the-Week Effects

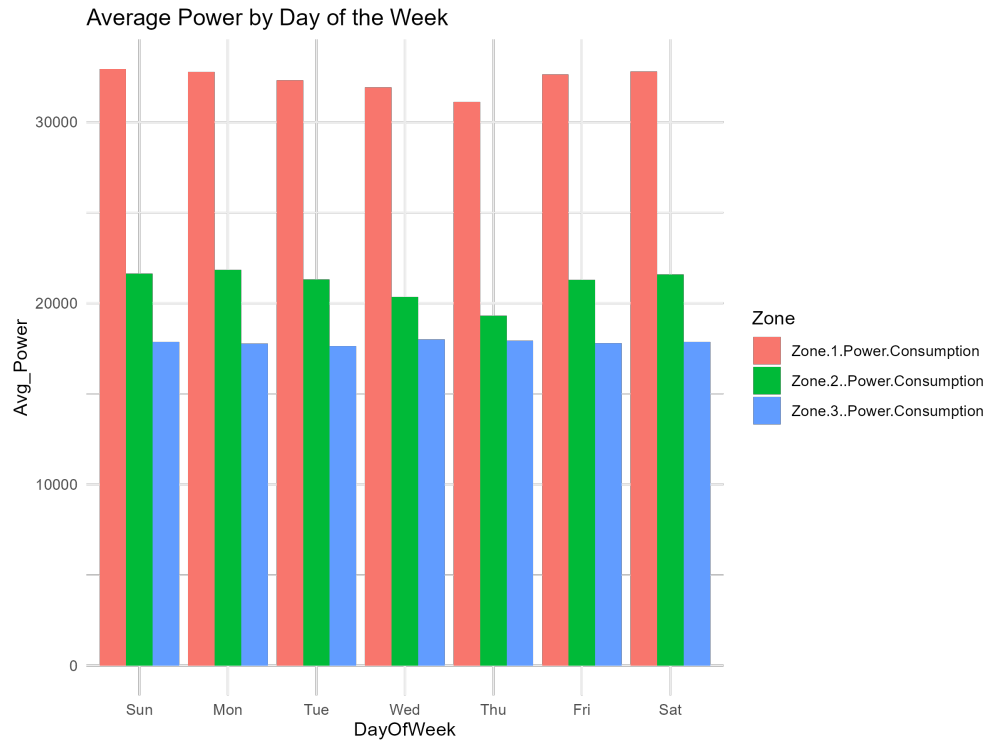


Figure 4: Average Power Consumption by Day of the Week

Figure 4 shows the average usage per day of the week. In Figure 5, where a smoothed time series compares the consumption on weekdays and weekends. It can be clearly seen that the weekend energy consumption is higher in zone 1 and zone 2. Zone 3 doesn't show significant difference.

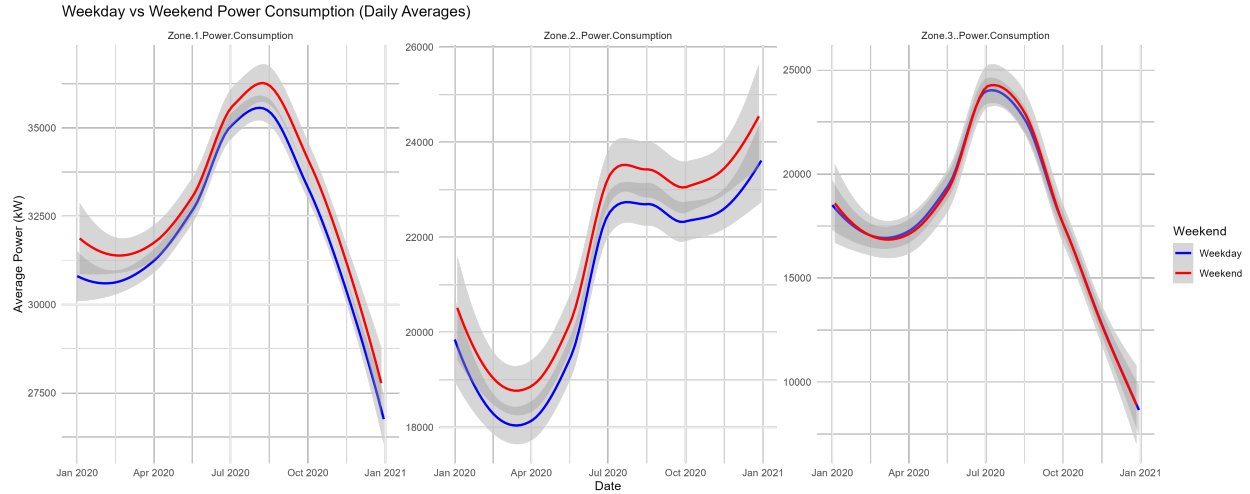


Figure 5: Weekday vs. Weekend Power Consumption (Daily Averages)

3.5 Outlier Analysis

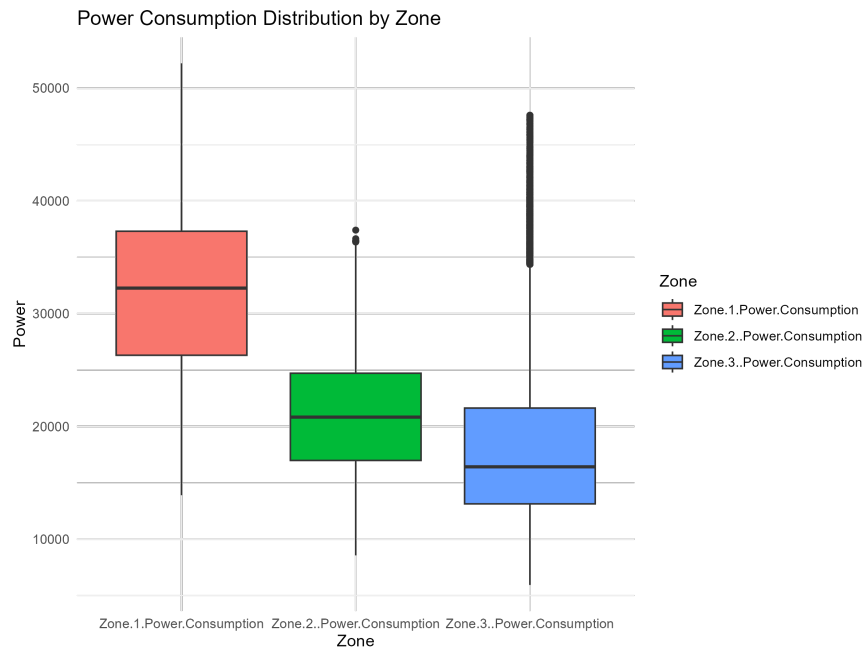


Figure 6: Boxplot for Outlier Detection

To identify abnormal power usage events, the IQR method is used to detect the outliers. Figure 6 presents the distribution of power values with potential outliers. The outliers clearly visible, particularly in Zone 3.

A summary of outlier statistics for each zone is provided in Table 2. The outlier analysis of power consumption across the three zones reveals meaningful differences in usage stability and variability. Zone 1 shows no outliers, suggesting a highly stable and predictable consumption pattern—possibly due to tightly regulated or uniform usage. Zone 2, with only 7 outliers out of over 52,000 data points, also demonstrates a high level of consistency, though the presence of a few extreme values hints at occasional deviations from the norm. In contrast, Zone 3 stands out with 1,191 outliers, comprising about 2.27% of its data.

Table 2: Summary of Outliers Detected by Zone

Zone	Total Points	Outlier Count	Outlier %
Zone 1	52,416	0	0.0%
Zone 2	52,416	7	0.013%
Zone 3	52,416	1,191	2.27%

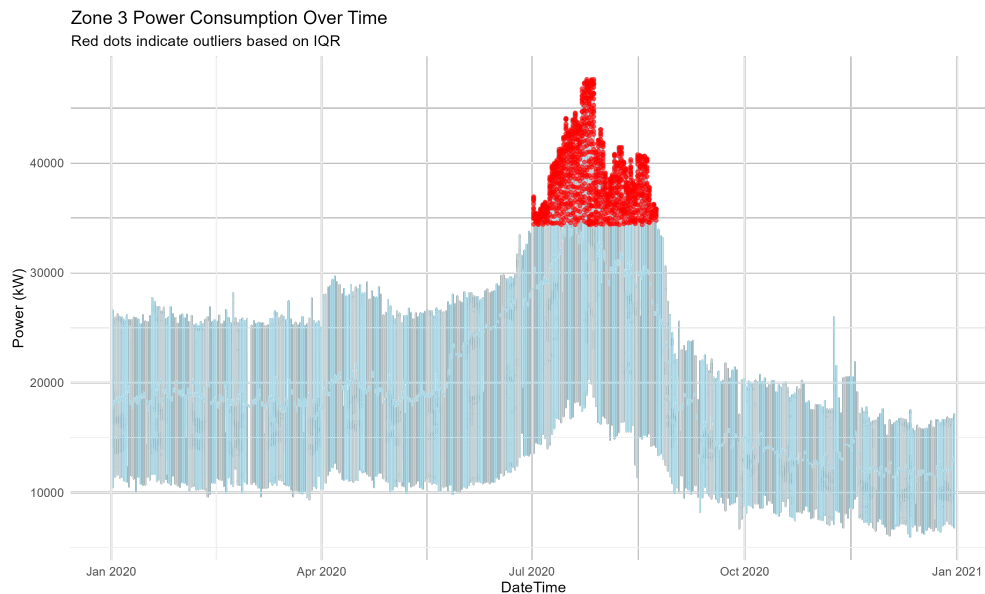


Figure 7: Zone 3 Power Consumption Over Time (Red Dots: Outliers)

The presence of outliers in zone 3 is highlighted here. In the summer months the energy usage in this zone is much higher. This should be further investigated. This comparatively high percentage indicates a more volatile consumption pattern. The variability could stem from irregular high-load usage, behavioral or operational anomalies, sensor errors, or natural fluctuations in demand due to external influences like weather or seasonality. While outliers are typically scrutinized or removed in some analyses, in this context, they offer valuable insights into zone-specific dynamics and warrant further investigation rather than outright exclusion.

3.6 Correlation with Scaled Predictors

Finally, the computed correlations between power and scaled weather variables (temperature, humidity, wind speed). Figure 8 show moderate-to-strong relationships between weather and consumption, reinforcing the potential for weather-based forecasting models. This helps in using the predictors in our model.

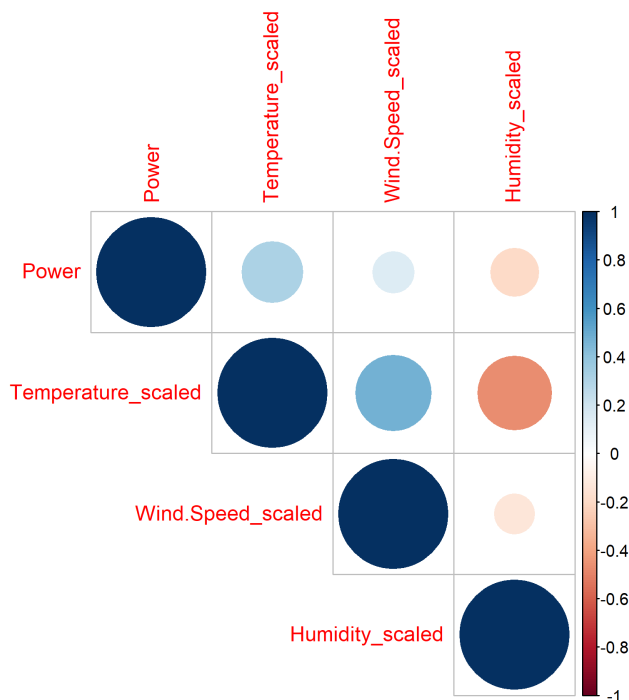


Figure 8: Correlation between Power and Scaled Predictors

4 Modeling Approach

4.1 Bayesian Framework

Energy usage patterns are not the same across every neighborhood, and that's where traditional models often fall short. A hierarchical Bayesian model will allow to capture the differences between zones while still learning from their similarities. Instead of treating each zone in isolation or assuming they all behave the same, this approach balances both perspectives. It also gives a clearer picture of uncertainty, meaning we can be more confident in the

predictions we make. By modeling each zone individually but within a shared framework, it will produce more reliable and insightful results, especially when the data for one zone might be limited or noisy. This makes the model not just accurate, but also adaptable and realistic.

The model incorporates a random intercept for each zone, acknowledging that baseline power consumption can vary by location. This hierarchical structure assumes that the zone-level intercepts are drawn from a common distribution, allowing for partial pooling. It balances between completely separate models for each zone (no pooling) and a single model for all zones (complete pooling), enabling the model to borrow strength across groups.

The priors used are weakly informative priors to regularize estimation while allowing the data to speak:

- Fixed effects: $\text{Normal}(0, 1)$ — centers estimates around 0 with moderate spread.
- Intercept: $\text{Normal}(0, 1)$ — reflects a similar expectation for the baseline.
- Random effects standard deviation: $\text{Cauchy}(0, 1)$ — appropriate for scale parameters, allowing for heavier tails and greater flexibility.
- Residual standard deviation: $\text{Cauchy}(0, 1)$ — allows for uncertainty in residual noise.

4.2 Model Specification

- Scaling of predictors and response: To facilitate convergence and interpretation, continuous variables (Temperature, Humidity, Wind Speed, general diffuse flows, and diffuse flows) were standardized. The response variable (Power consumption) was also scaled to have mean zero and unit variance.
- Choice of likelihood (Gaussian): Given that the scaled power consumption data appeared approximately normally distributed, a Gaussian likelihood was appropriate for modeling the continuous response variable.
- Mathematical Model Specification:

Hierarchical Model Equation

$$\text{Power}_{ij} = \beta_0 + u_{0j} + \beta_1 \cdot \text{Temperature}_{ij} + \beta_2 \cdot \text{Humidity}_{ij} + \beta_3 \cdot \text{Month}_{ij} + \beta_4 \cdot \text{Weekend}_{ij} + \varepsilon_{ij}$$

(The features power, temperature and humidity are scaled. Month and weekend are encoded.)

- i indexes observations (time points), and j indexes zones (Zone 1, 2, or 3).
- β_0 is the global intercept (overall baseline).
- $u_{0j} \sim \mathcal{N}(0, \sigma_{\text{zone}}^2)$ is the random intercept for zone j , capturing deviation from the global intercept.

- $\beta_1, \beta_2, \beta_3, \beta_4$ are fixed effect coefficients for scaled predictors: temperature, humidity, month (numerical), and weekend indicator.
- $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ is the residual error term.

This hierarchical formulation allows the model to borrow strength across different zones while accounting for zone-specific variability in baseline power consumption.

4.3 Implementation

The hierarchical model was implemented in R using the `brms` package, which relies on Stan as the probabilistic programming backend. For Bayesian inference, the model was estimated using Markov Chain Monte Carlo (MCMC) sampling with four chains, each running for 4,000 iterations, including 1,000 warm-up iterations, yielding a total of 12,000 post warmup draws. The model exhibited good convergence, with no divergent transitions or sampling warnings. As a result, the default values for tuning parameters such as `adapt_delta` and `max_treedepth` were maintained.

5 Results

Table 3: Posterior Summary of Model Parameters

Parameter	Estimate	Est. Error	l-95% CI	u-95% CI	Rhat	Bulk ESS	Tail ESS
<i>Fixed Effects</i>							
Intercept	0.17	0.50	-0.84	1.21	1.00	1445	2558
Temperature_scaled	0.32	0.00	0.31	0.32	1.00	11376	6869
Humidity_scaled	-0.05	0.00	-0.05	-0.04	1.00	12391	8834
Month_num	-0.03	0.00	-0.03	-0.02	1.00	12314	7537
Weekend	0.05	0.00	0.04	0.05	1.00	7380	6592
<i>Group-Level Effects (Zone)</i>							
sd(Intercept)	1.05	0.52	0.44	2.45	1.00	1595	2656
<i>Distributional Parameter</i>							
sigma	0.64	0.00	0.63	0.64	1.00	14282	8105

- Posterior summaries for coefficients:

- Temperature has a positive effect on power consumption ($\hat{\beta} = 0.32$).
- Humidity has a small negative effect ($\hat{\beta} = -0.05$).
- Later months are associated with slightly lower power usage ($\hat{\beta} = -0.03$).
- Weekends show a small positive effect on consumption ($\hat{\beta} = 0.05$).

- Interpretation of key parameters:

A one standard deviation increase in temperature leads to a 0.32 standard deviation increase in power usage, suggesting strong reliance on heating/cooling systems. The weekend effect, though modest, could indicate differences in residential versus commercial energy usage patterns.

- Random effects across zones:

Table 4: Zone-Specific Random Intercepts (u_{0j})

Zone	Estimate	Est. Error	2.5% Quantile	97.5% Quantile
Zone1	0.9491	0.4982	-0.0939	1.9652
Zone2	-0.3200	0.4982	-1.3616	0.6947
Zone3	-0.6802	0.4982	-1.7226	0.3357

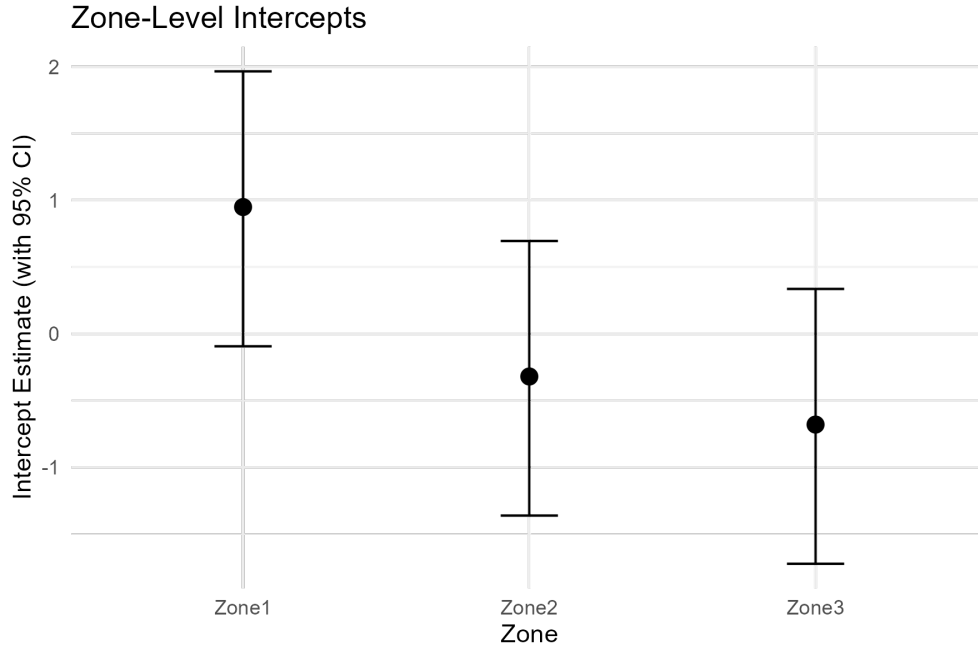


Figure 9: Zone level intercepts

Table 5: Combined Fixed and Random Intercepts ($\beta_0 + u_{0j}$)

Zone	Estimate	Est. Error	2.5% Quantile	97.5% Quantile
Zone1	1.1188	0.6679	0.0758	2.1349
Zone2	-0.1503	0.6679	-1.1919	0.8644
Zone3	-0.5105	0.6679	-1.5529	0.5054

There was substantial variation between zones, with the standard deviation of the intercepts estimated at 1.05 (95% CI: 0.44 to 2.45). Zone-level intercepts indicate that Zone 1 has higher baseline consumption than the others.

6 Model Diagnostics and Evaluation

- The trace plots demonstrated good mixing across all chains, indicating that the MCMC sampling converged effectively for every parameter as shown in the Figures 10 and 11.

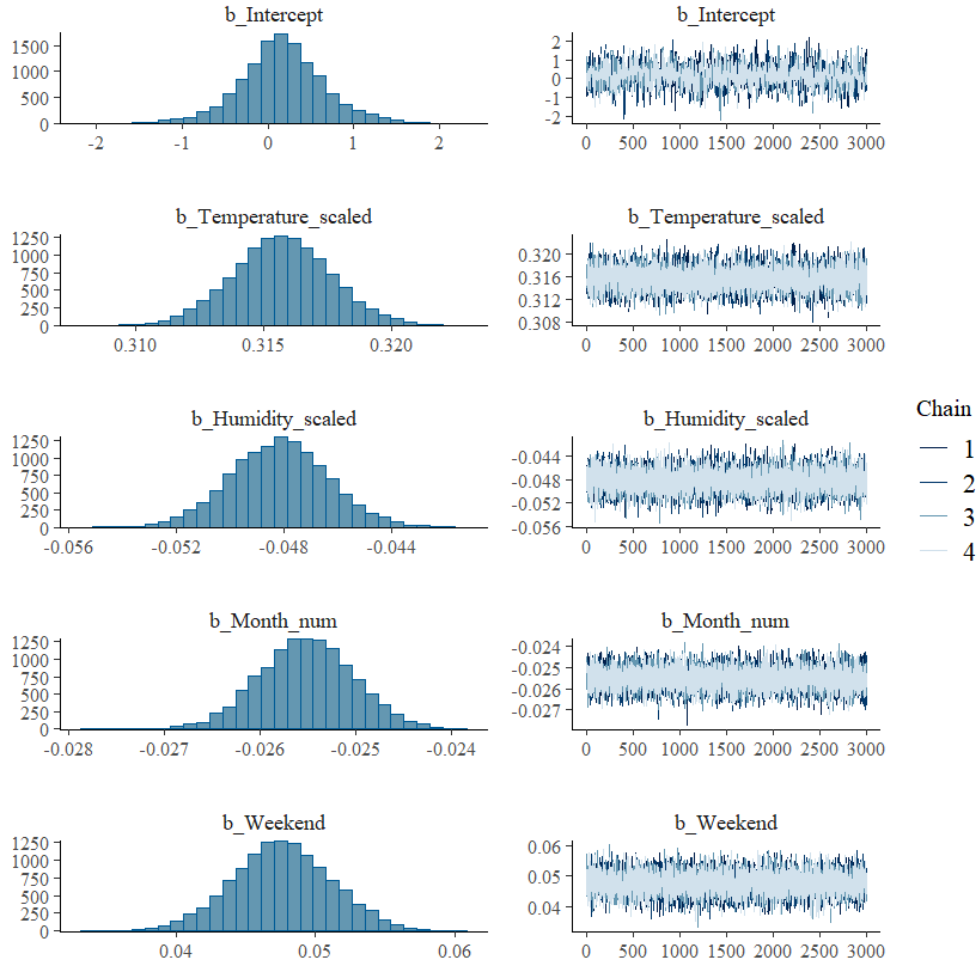


Figure 10: Trace plots

- The potential scale reduction factor, $\hat{R}$, was consistently equal to 1.00 for all parameters, confirming convergence.
- The effective sample sizes (ESS) for both the bulk and tail regions were sufficiently large—for example, Temperature had ESS values of 11,376 and 6,869—ensuring that the posterior estimates are reliable.
- There were no divergent transitions detected during sampling, which suggests the posterior distribution was well-behaved and the sampler performed efficiently.

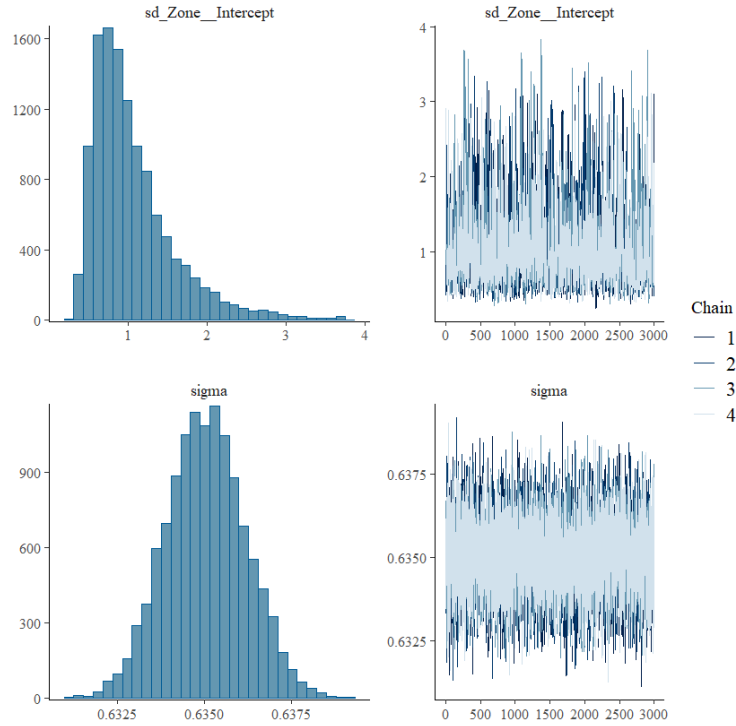


Figure 11: Trace plots

- Posterior predictive checks visually confirmed that the model fits the observed data well, accurately capturing its overall distribution and variability.

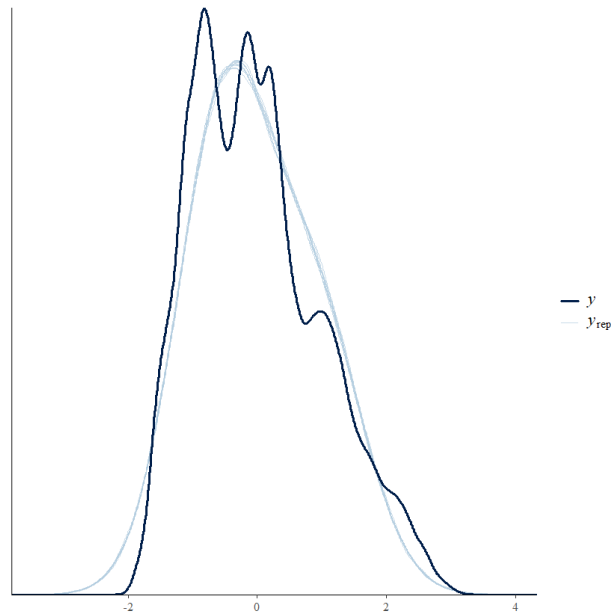


Figure 12: Posterior Predictive Probability

7 Limitations and Scope of Future Work

Some potentially relevant predictors (e.g., wind speed, solar radiation, time of day) were not included in the model. The model doesn't account for potential autocorrelation or temporal structure in the data.

Future iterations could include additional weather variables, interaction effects, or time series structure. Modeling with splines or Gaussian processes might capture nonlinear trends. Moreover, introducing varying slopes per zone could allow for more nuanced modeling of zone-specific responses to predictors.

8 Conclusion

Temperature is the dominant predictor of power consumption, with additional effects from humidity and calendar-related features. There are substantial differences in baseline usage across zones that are effectively captured through the hierarchical structure.

The multilevel Bayesian approach allows for flexible, robust modeling that accounts for group-level variability while pooling information efficiently. This is particularly valuable when data are unbalanced or when making inferences at both individual and group levels.

Acknowledgments

Salam, A., & El Hibaoui, A. (2018, December). *Comparison of Machine Learning Algorithms for the Power Consumption Prediction: Case Study of Tetouan City*. In *2018 6th International Renewable and Sustainable Energy Conference (IRSEC)* (pp. 1–5). IEEE.